

Daily Problem: 12–Feb–2026

Problem Statement

Let $L \subseteq \{a\}^*$ be any language over the singleton alphabet $\Sigma = \{a\}$ (not necessarily regular). Show that L^* is regular.

Proof

Every word over $\{a\}$ has the form a^n for a unique $n \in \mathbb{N}$. Hence we can identify L with the set of lengths

$$S = \{n \in \mathbb{N} : a^n \in L\}.$$

Under this identification, the Kleene star L^* corresponds to the set of all finite sums of elements of S :

$$a^t \in L^* \iff t = n_1 + \cdots + n_k \text{ for some } k \geq 0 \text{ and } n_i \in S.$$

(Here $k = 0$ gives the empty sum 0, corresponding to $\varepsilon = a^0 \in L^*$.)

We now show that the set of lengths of words in L^* is ultimately periodic, hence L^* is regular.

Step 1: If $L \subseteq \{\varepsilon\}$, then L^* is trivially regular

If $S \subseteq \{0\}$, then $L \subseteq \{\varepsilon\}$. In this case $L^* = \{\varepsilon\}$, which is regular. So assume from now on that S contains some positive integer.

Step 2: Let d be the gcd of the positive lengths in L

Let

$$S^+ = S \cap \{1, 2, 3, \dots\}.$$

Assume $S^+ \neq \emptyset$ and define

$$d = \gcd(S^+).$$

Then every element of S^+ is divisible by d , and therefore every finite sum of elements of S^+ is divisible by d as well. Hence every word in L^* has length divisible by d , except possibly ε which has length 0 and is also divisible by d . So we have the inclusion

$$L^* \subseteq \{a^{dn} : n \geq 0\}.$$

Step 3: Eventually, every multiple of d belongs to L^*

Choose finitely many elements of S^+ ,

$$\ell_1, \ell_2, \dots, \ell_m \in S^+,$$

such that

$$\gcd(\ell_1, \dots, \ell_m) = d.$$

(This is possible because gcd is determined by a finite subset.)

Define scaled lengths

$$t_i = \ell_i/d \in \mathbb{N}, \quad 1 \leq i \leq m,$$

so that

$$\gcd(t_1, \dots, t_m) = 1.$$

Now consider the numerical semigroup

$$H = \left\{ \sum_{i=1}^m x_i t_i \mid x_i \in \mathbb{N} \right\}.$$

A standard fact from elementary number theory (the “coin problem” / Frobenius-type phenomenon) says:

If $\gcd(t_1, \dots, t_m) = 1$, then there exists N_0 such that every integer $n \geq N_0$ lies in H .

Equivalently, for all $n \geq N_0$, we can write

$$n = x_1 t_1 + \cdots + x_m t_m$$

for some $x_i \in \mathbb{N}$. Multiplying by d ,

$$dn = x_1 \ell_1 + \cdots + x_m \ell_m.$$

But $\ell_i \in S$ means $a^{\ell_i} \in L$, and concatenating x_i copies of a^{ℓ_i} gives a word in L^* of length $x_i \ell_i$. Thus the concatenation of all these blocks lies in L^* and has total length dn . Therefore,

$$a^{dn} \in L^* \quad \text{for all } n \geq N_0.$$

So there exists a cutoff N_0 such that:

$$L^* \cap \{a^{dn} : n \geq N_0\} = \{a^{dn} : n \geq N_0\}.$$

Step 4: A regular expression for L^*

From the previous steps, L^* consists of:

- finitely many exceptional words of lengths $< dN_0$, and
- all words whose length is a multiple of d and at least dN_0 .

Let

$$F = \{a^n \in L^* : n < dN_0\}.$$

This is a finite language, hence regular.

Also,

$$\{a^{dn} : n \geq N_0\} = a^{dN_0} (a^d)^*,$$

which is regular.

Therefore,

$$L^* = F \cup a^{dN_0} (a^d)^*,$$

a union of regular languages, hence regular.

Conclusion

For any language $L \subseteq \{a\}^*$, the Kleene star L^* is regular. In fact, L^* is ultimately periodic in length: beyond some point it contains exactly all words whose length is divisible by a fixed d .